

RESEARCH ARTICLE

WILEY

Underwater image enhancement method based on entropy weight fusion

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Abstract

Due to the absorption and scattering of underwater light, underwater images often have some phenomena such as colorcast, low contrast, and blurred details. Therefore, an underwater image enhancement method based on entropy weight fusion is proposed. First, white balance processing can effectively correct the blue (green) color appearance of the image. Then the white-balanced images are converted from RGB space to LAB space, and L channel is processed with improved adaptive gamma correction, and then converted back to RGB space. CLAHE and bilateral filtering are performed in RGB space. The RGB space is converted to HSV space, the V channel is processed by single-scale retinex algorithm combined with guided filtering, stretching the R channel with s-cosine curve, and then converted back to RGB space. Finally, the three results are fused by entropy weight to obtain the final enhanced image. Experimental results show that the proposed algorithm can improve the contrast and clarity of underwater images, and effectively remove colorcast.

KEYWORDS

color compensation, entropy weight method, image enhancement, retinex algorithm, underwater image

1 | INTRODUCTION

Countries around the world are increasingly demanding energy sources such as oil and natural gas. Due to the shortage of land resources, people pay more and more attention to the development of marine alternative resources.¹ In the development of marine resources, the acquisition of deep-sea images is very important, but the deep-sea images are often blurred. Therefore, deep-sea image processing technology plays an important role in the acquisition of marine resources. Water medium and suspended particles can absorb and attenuate light, respectively, and the longer the wavelength of light is, the easier it is to be absorbed. The wavelength of light gradually becomes shorter in the visible band from red to purple,² so red light is more easily absorbed, and underwater images tend to appear blue and green. In addition to colorcast, because the light is affected by the attenuation of underwater suspended particles, underwater images are generally fuzzy and poor visibility, so it is of great practical significance to improve the clarity of underwater images.

1.1 | Related works

The current methods for underwater image sharpening mainly include image enhancement, image restoration, and deep learning. In this part, we will summarize from these three aspects.

The image enhancement method achieves the image enhancement effect by adjusting the gray value of pixels, and does not consider the physical imaging model of underwater images. Ancuti et al.³ used white balance and local adaptive histogram equalization of classical contrast to obtain clear shallow images through multi-scale fusion. Shen et al.⁴ proposed an image clearness algorithm for $l\alpha\beta$ color space, which combined bilateral filtering and tetrolet transform to significantly improve image color and detail. Guo et al.⁵ proposed a simple and effective low-light level image enhancement method. The illumination of each pixel is estimated by finding the maximum values in the R, G, and B channels, respectively. Shi et al.⁶ proposed an underwater grayscale image enhancement method based on controllable kernel function bilateral filtering retinex algorithm, which has the function of correcting colorcast. Fu et al.⁷ proposed a method to adjust the illumination by fusing multiple derivatives of the initial estimated illumination map, which enhanced the images under different low-light conditions. Iqbal et al.⁸ proposed an unsupervised color correction method to enhance underwater images. Fu et al.⁹ proposed an underwater image enhancement method based on retinex to enhance contrast and correct colorcast of a single image.

In the image restoration method, the image model of underwater image is established and the clear image is obtained by using the restoration parameters. Yang et al.¹⁰ proposed a sea-level attenuation recovery method based on statistical prior, which is suitable for underwater fast processing. Galdran et al.¹¹ proposed the dark channel prior method of red channel inversion, which can effectively improve the image restoration quality. Li et al.¹² proposed guiding triangular filtering to solve the problem of uneven lighting of artificial light source. Emberton et al.¹³ proposed the method of stratified estimation of atmospheric light to defogging underwater images. Akkaynak et al.¹⁴ proposed an improved underwater imaging model method, which has a good correction effect on complex underwater scenes. Hu et al.¹⁵ proposed a target brightness inversion method based on the estimation of polarization degree and the backscattering intensity at different positions of the image. Peng et al.¹⁶ added adaptive color correction to the image formation model to eliminate colorcast and improve contrast. Song et al.¹⁷ proposed an underwater image enhancement method based on the statistical model of background light and projection image optimization.

With the rapid development of deep learning, it is gradually applied to visual tasks. In recent years, deep learning-based methods have been widely used in underwater image enhancement. Li et al.¹⁸ proposed an unsupervised generation network method for real-time color correction of underwater images, and the image quality was effectively improved. Anwar et al.¹⁹ discussed the deep learning method for underwater images in detail. Chen et al.²⁰ proposed a real-time and adaptive underwater image processing method based on generative adversarial network. Fu et al.²¹ designed a compressed histogram equalization method to supplement the data-driven deep learning method. The proposed deep network has a lightweight structure, which can effectively solve the problems of colorcast and contrast reduction of underwater images. Li et al.²² proposed an underwater image enhanced convolutional neural network model based on underwater scene prior, which can directly reconstruct clear underwater images without estimating the parameters of the underwater imaging model. Islam et al.²³ proposed a real-time underwater image enhancement model based on conditional generation adversary-network, and constructed a large scale data set composed of paired and unpaired.

1.2 | Contributions

Some progress has been made in image enhancement, but the effect on contrast, colorcast and brightness of deep-sea image enhancement is not good. There are few underwater ground-truth images based on deep learning method, and there is a lack of reference images in the model training process, and the training is time-consuming. In this article, a new fusion framework is proposed to solve these problems. For a single underwater image, no reference image is required, which can effectively improve the visual effect of the underwater image and take a short time. The algorithm flow chart of this article is shown in Figure 1. First, the loss of red channel is compensated and colorcast of underwater image is corrected by white balance processing. Then, the color offset corrected images are respectively processed as follows: RGB space is converted into LAB space, L channel is processed by improved adaptive gamma correction and then converted back to RGB space to obtain input 1; input two is obtained by CLAHE and bilateral filtering in RGB space. The RGB space is converted to HSV space, the V channel is processed by single-scale Retinex, stretching the R channel with s-cosine curve, and then converted back to RGB space to obtain input three. Finally, the three results are fused with entropy weight method to obtain the enhanced image. The main contributions of our article are summarized as follows:

1. We propose a simple fusion framework in three color channels, which can effectively blend images with different dominant features after processing a single image to enhance a single input underwater image.

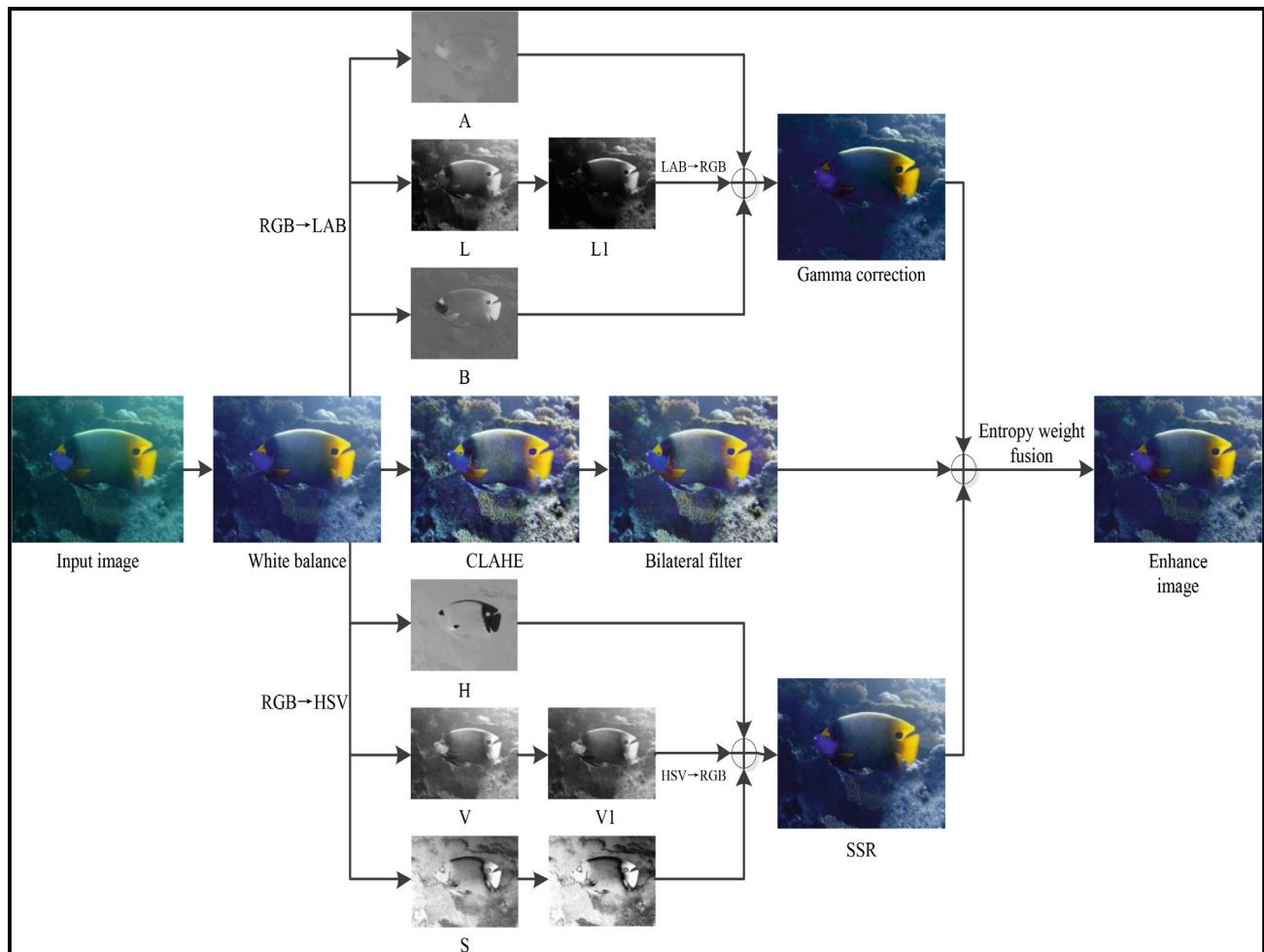


FIGURE 1 Flow chart of the proposed method

2. We propose an improved adaptive gamma correction method to effectively solve the problem that the effect of manually setting gamma parameters for different images is not ideal, and to correct the too bright images processed by the white balance method to improve the clarity of the images.

3. We propose an adaptive weight selection method for underwater images with different processing results, which is objective.

The rest of the article is structured as follows. The second part introduces the algorithm in this article, and the third part compares the qualitative and quantitative evaluation of this method.

2 | PROPOSED METHOD

2.1 | Color correction

The purpose of white balance is to enhance the appearance of an image by eliminating or mitigating changes in the color of an object caused by the projection of light. In deep water, the attenuation characteristic of water medium to light is that the long wave attenuation is more serious than the short wave attenuation. As a result, deep-sea images tend to have a blue (green) appearance. Among the color channels of underwater images, the information of the green channel is better preserved than that of the red and blue channels. Therefore, the green channel can be used to compensate for the red channel. Gray world is assumed that the R, G, and B components of the color image tend to the same gray value, and the average reflection in the scene is fixed. The color distribution of the light source in each channel is estimated by the

average value of each channel. Therefore, red and green light attenuation is unbalanced, and compensation should be proportional to the difference between the mean of the green channel and the mean of the red channel. The gray world method is adopted to remove the blue (green) appearance, but there will be serious red artifacts, so the loss of the red channel should be compensated first.²⁴

At each pixel position (x) compensation n red channel I_{rc} is defined as:

$$I_{rc}(x) = I_r(x) + \alpha \cdot (\bar{I}_g - \bar{I}_r) \cdot (1 - I_r(x)) \cdot I_g(x), \quad (1)$$

where $I_1 = cI_1'$, $I_1 = cI_1'$ represents the red channel and green channel of image $I_1 = cI_1'$, and is normalized for each channel, and each channel is in the range of $[0, 1]$. $I_1 = cI_1'$ and $I_1 = cI_1'$ are the mean of $I_1 = cI_1'$ and $I_1 = cI_1'$, respectively. $I_1 = cI_1'$ represents a constant parameter, in this article $I_1 = cI_1'$. After the red channel compensation, the traditional gray world hypothesis is used to estimate and compensate colorcast of the light source.

2.2 | Improved adaptive gamma correction

White balance underwater images tend to be too bright, and gamma correction is used to increase the difference between light and dark images.²⁴ Gamma correction can enhance the proportion of the region of interest in the image, with good dynamic range compression effect, and adjust the overall brightness of the image by using two variable parameters $I_1 = cI_1'$ and $I_1 = cI_1'$. The gamma transformation is expressed as follows,

$$I_1 = cI_1', \quad (2)$$

where $\gamma = \exp\left(\frac{\text{Var}(I_L) - D\text{Mean}(I_L)}{\text{Var}(I_L)}\right)$ represents the intensity of the input image, $\gamma = \exp\left(\frac{\text{Var}(I_L) - D\text{Mean}(I_L)}{\text{Var}(I_L)}\right)$ represents the intensity of the output image, both $\gamma = \exp\left(\frac{\text{Var}(I_L) - D\text{Mean}(I_L)}{\text{Var}(I_L)}\right)$ and $\gamma = \exp\left(\frac{\text{Var}(I_L) - D\text{Mean}(I_L)}{\text{Var}(I_L)}\right)$ can adjust the shape of the gamma function.

Different image quality is different, the effect of setting the gamma transform parameters manually is not ideal. We propose an improved adaptive gamma correction method to transform the white balanced image from RGB color space to LAB color space. Under the premise of keeping the color component A, B, and invariant, the adaptive gamma transformation is carried out for the brightness component L. The improved adaptive gamma parameters are shown as follows,

$$\gamma = \exp\left(\frac{\text{Var}(I_L) - D\text{Mean}(I_L)}{\text{Var}(I_L)}\right), \quad (3)$$

where I_L represents the L channel of image I , and Var and Mean represent the variance and mean of I_L , respectively. Experiments show that the effect is better in $D = 15$. In this article, $c = 1$.

In order to keep the color after color correction unchanged, the image was converted to the LAB color space, the color component A, B and remained unchanged, and only the brightness component L was processed, so as to obtain the image conforming to human physiological characteristics. The brightness component L after gamma transformation is combined with the color component A, B and converted to the RGB color space to realize the brightness adjustment of the underwater image. Mark the location of the red box as shown in Figure 2, as shown in Figure 2a is the processed image is white balance, image too bright, Figure 2b is the result of the white balance image using gamma correction, the correction of a bright white balance image, Figure 2c is the result of using adaptive gamma correction improved in this article for white-balanced images, relatively primitive gamma correction, the improved algorithm not only corrects the over-brightness of the white balance image, but also provides richer details and better contrast enhancement.

2.3 | Limiting contrast adaptive histogram equalization and bilateral filtering

The image still has low contrast and noise after color correction. The enhanced image with uniform brightness is obtained by the limited contrast adaptive histogram equalization algorithm.²⁵

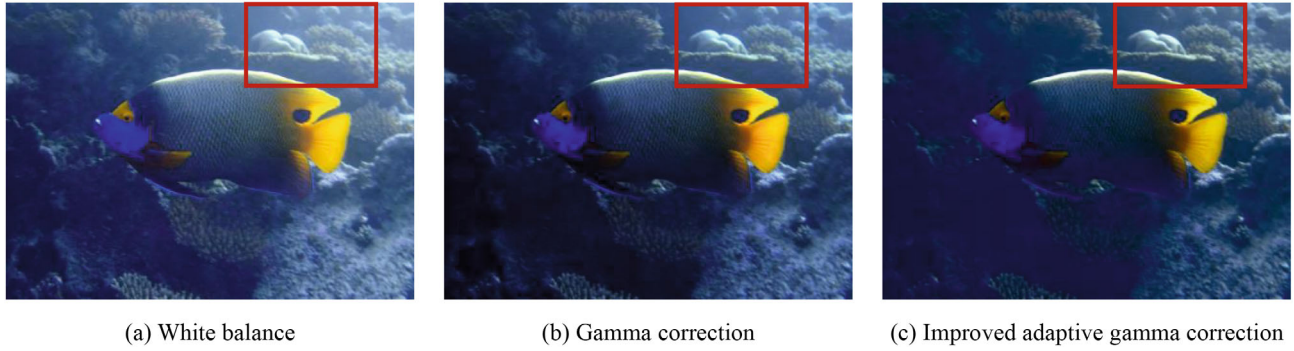


FIGURE 2 Brightness adjustment of image

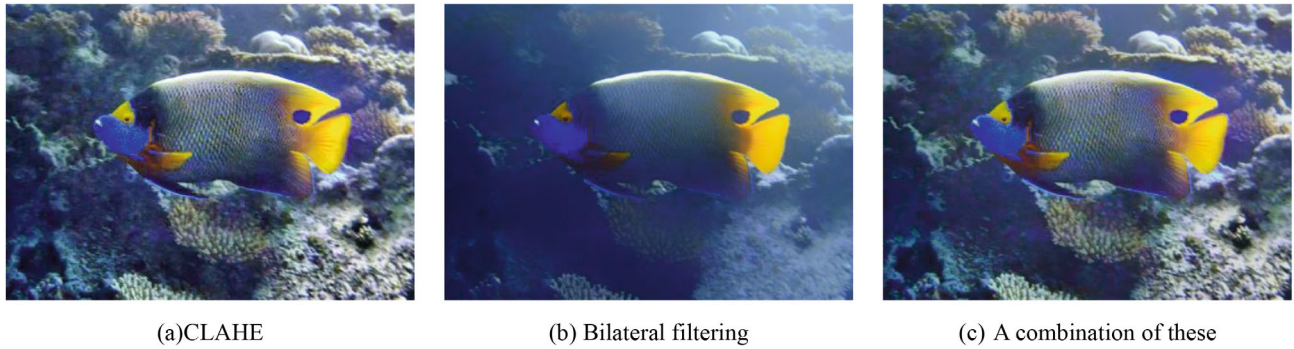


FIGURE 3 Contrast adjustment of image

Because of the special underwater lighting conditions, it is very important to remove image noise. Bilateral filtering is a nonlinear filtering method which combines spatial proximity and pixel value approximation of images and considers spatial information and gray level similarity at the same time. Bilateral filter³ is used to remove noise and maintain edges to improve image clarity. Wiener filtering and Gaussian filtering will obviously blur edges. Specific definitions are as follows:

$$h(x) = \frac{1}{k_r} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\xi) s(f(\xi), f(x)) d\xi, \quad (4)$$

$$h(x) = \frac{1}{k_d} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\xi) c(\xi, x) d\xi, \quad (5)$$

where $s(f(\xi), f(x))$ is brightness similarity, k_r and k_d are normalized coefficients, $c(\xi, x)$ is the Euclidean distance between ξ and x .

The transfer function of bilateral filter is expressed as:

$$BF_i(x, y) = \frac{1}{k(x)} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(\xi) c(\xi, x) s(f(\xi), f(x)) d\xi, \quad (6)$$

where $BF_i(x, y)$ is the input function, $BF'_i(x, y)$ is the output function. The discrete form of a bilateral filter is defined as follows:

$$BF'_i(x, y) = \sum_{k=-q}^q \sum_{l=-q}^q A(k, l, x, y) BF_i(x - k, y - l), \quad (7)$$

where $A(k, l, x, y)$ is the Gaussian function, the definition is as follows:

$$A(k, l, x, y) = \frac{\exp\left(-\frac{k^2+l^2}{2\sigma_d^2} - \frac{(R(x,y)-R(x-k,y-l))^2}{2\sigma_r^2}\right)}{C(x, y)}, \quad (8)$$

where (x, y) is the current pixel, (k, l) is the adjacent pixel, $R(x, y)$ is the corresponding pixel value, σ_d is the spatial domain filter standard deviation, σ_r is the range filter standard deviation.

CLAHE and bilateral filtering are applied to the color corrected images in the RGB color space, and the overall good effect of contrast enhancement and noise removal can be obtained. Figure 3 is the algorithm comparison diagram of adaptive histogram equalization with limited contrast, bilateral filtering and the combination of the two based on the results of white balance. It can be seen from Figure 3 that the image obtained by combining the two methods is better in terms of brightness and denoising, the image is more natural, and this result is taken as the second input.

2.4 | Retinex algorithm

Retinex²⁶ is based on the theory that the color of an object is determined by its ability to reflect light, rather than by the absolute value of the reflected light intensity. The retinex algorithm's scale changes with the variation of Gaussian scale, while the Gaussian scale does not change in retinex algorithm, which is called single-scale retinex algorithm, referred to as SSR.²⁷

Due to the characteristics of the Gaussian function selected in SSR, the enhanced image cannot guarantee the detail information and color information at the same time.²⁸ Aiming at the problem of detail keeping in SSR, this article uses guided filter instead of Gaussian filter. The guided filter is a local linear smoothing filter that can preserve the edge.²⁹ The input and output of the guiding filter function satisfy the following linear relation:

$$q_i = a_k I_i + b_k, \forall i \in w_k, \quad (9)$$

where q is the output image, I is the guide image, i and k are the pixel index, w is the window of the image, a and b are the coefficients of the function when the center of w is at the position of k . We can use the input image or other suitable image as the guide image. The derivative of both sides of the above equation is:

$$\nabla q = a \nabla I. \quad (10)$$

Therefore, when the guided image I has a gradient, the output image q also has a corresponding gradient. So the guided filtering can maintain the edge features of the image.

To obtain the coefficient of the linear function, the cost function of the small difference between the output image q and the real image p of the fitting function is:

$$E(a_k, b_k) = \sum_{i \in w_k} ((a_k I_i + b_k - p_i)^2 + \epsilon a_k^2). \quad (11)$$

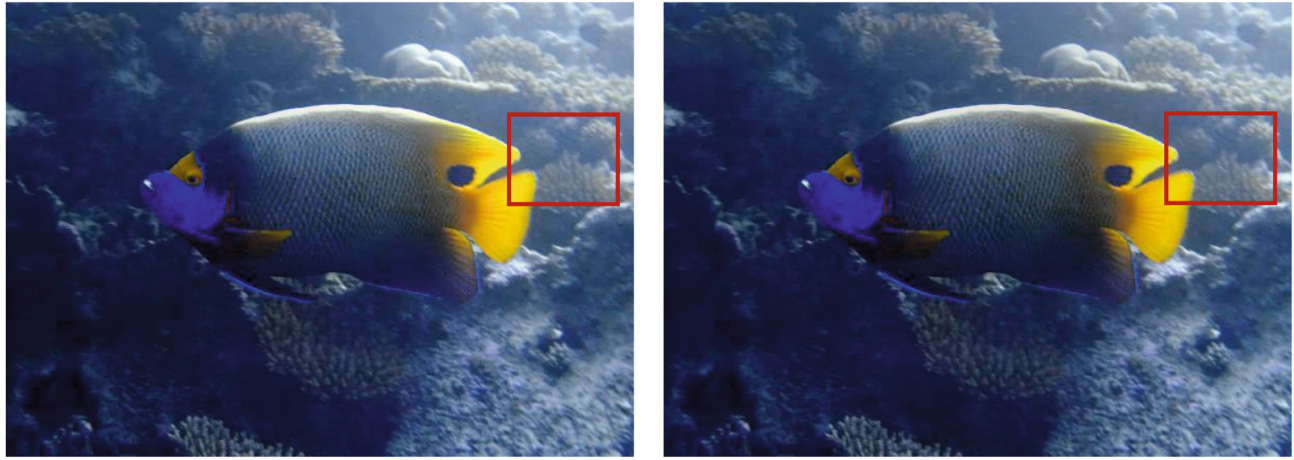
According to the least square method:

$$a_k = \frac{\frac{1}{|w|} \sum_{i \in w_k} I_i p_i - u_k \bar{p}_k}{\sigma_k^2 + \epsilon},$$

$$b_k = \bar{p}_k - a_k \mu_k, \quad (12)$$

where u_k represents the average value of image I , σ_k^2 represents the variance of image I , $|w|$ is the number of pixels in w_k , and $\bar{p}_k$ is the mean value of image p .

The HSV color space is more consistent with the visual characteristics of human eyes. The underwater image is converted to the HSV space, and the hue H of the color corrected image is kept unchanged. Stretching the R channel with



(a) Gaussian filter

(b) Guide the filtering

FIGURE 4 Contrast image of algorithm

s-cosine curve³⁰ to compress the regions with high and low gray values of the image to a certain extent, and stretch the intermediate gray values to expand the dynamic range of the image. Stretching function:

$$S'(x, y) = \left(-\frac{1}{2}\right) \times \cos(3 \times S(x, y) + 0.02) + 0.5, \quad (13)$$

where $S'(x, y)$ are the pixel value of the output image, $S(x, y)$ are the pixel value of the input image, which is normalized to the range of $[0, 1]$.

The brightness channel V was enhanced by SSR combined with guidance filter to get the image suitable for observation, and then the image was converted back to RGB space to get the enhanced image. As can be seen from the red box in Figure 4, the contrast and clarity of SSR with guided filtering are better than those with Gaussian filtering. Therefore, SSR with guided filtering is used as the third input.

2.5 | Entropy weight fusion

Originally a thermodynamic concept, entropy was introduced into information theory by Shannon in 1948. Entropy is used to describe the uncertainty of the signal source and is a measure of the disorder degree of the system. Entropy is a measure of the disorder degree of the system. The smaller the information entropy of the evaluation index, the greater the variation degree of the index value, the greater the information reflected, and the greater the corresponding weight. On the contrary, the weight is smaller.³¹ When the values of objects on a certain index are completely equal, the entropy value is the largest, so this index can be removed from the evaluation. Entropy weight method is an objective weight method based on original data, which is not easy to be affected by subjective factors.³² Therefore, the three inputs in this article use this method to determine the weight. The principle of entropy weight method is as follows:

There are m evaluation objects and n error indexes, and the index matrix is.

$$X' = \begin{bmatrix} x'_{11} & x'_{12} & \cdots & x'_{1n} \\ x'_{21} & x'_{22} & \cdots & x'_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ x'_{m1} & x'_{m2} & \cdots & x'_{mn} \end{bmatrix}. \quad (14)$$

Standardize X'

$$R = (r_{ij})_{m \times n}, \quad (15)$$

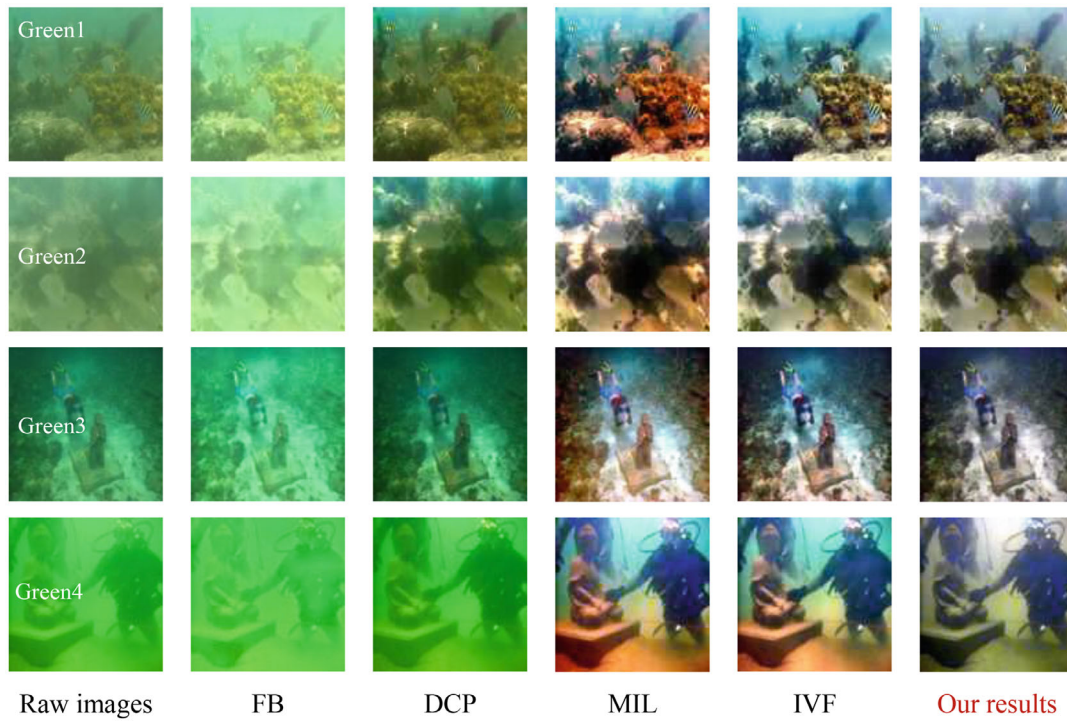


FIGURE 5 The greenish underwater images

where

$$r_{ij} = \frac{x_{ij}}{\sum_{i=1}^m x_{ij}}. \quad (16)$$

r_{ij} is the value of the j th error index on the i th evaluation object, $r_{ij} \in [0, 1]$.

The calculation formula of close to ideal scheme is as follows:

$$d_{ij} = D_{ij} / \sum_{i=1}^m \sum_{j=1}^n D_{ij}, \quad (17)$$

where

$$D_{ij} = \frac{r_j^*}{r_{ij}} = \frac{\min_i |r_{ij}|}{r_{ij}}. \quad (18)$$

D_{ij} is the expression form of evaluation index. The smaller the evaluation value, the better the effect.

The information entropy:

$$e_j = -K \sum_{i=1}^m r_{ij} \ln r_{ij}, \quad (19)$$

where $K = \frac{1}{\ln m}$, $r_{ij} = \frac{d_{ij}}{d_j}$, $d_j = \sum_{i=1}^m d_{ij}$, when $r_{ij} = 0$, let $r_{ij} \ln r_{ij} = 0$.

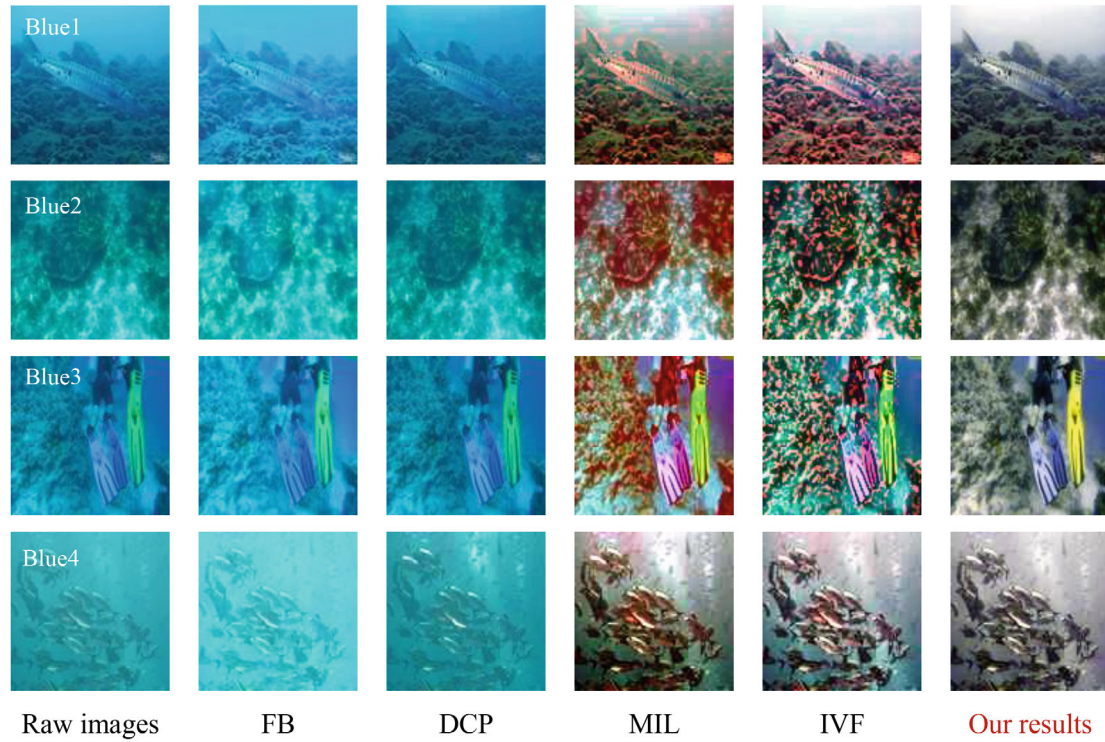


FIGURE 6 The bluish underwater images

The weight of the j th index is:

$$\theta_j = \frac{1 - e_j}{\sum_{j=1}^n (1 - e_j)}. \quad (20)$$

The weight calculation formula is as follows:

$$w_i = \sum_{j=1}^n \theta_j D_{ij} / \sum_{i=1}^m \sum_{j=1}^n \theta_j D_{ij}, \quad (21)$$

where $0 \leq w_i \leq 1$, $\sum_{i=1}^m w_i = 1$.

In this article, three inputs are taken as evaluation objects, UIQM, information entropy, and average gradient were used as evaluation indexes. The fusion image is:

$$I_o = w_1 I_1 + w_2 I_2 + w_3 I_3. \quad (22)$$

3 | EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we compare several existing classical and state-of-the-art methods with our proposed method to evaluate the performance of our proposed method. Comparative methods include fusion-based (FB),⁷ dark channel prior (DCP),³³ minimum information loss (MIL),^{34,35} image and videos by fusion (IVF).³ The underwater image data set used in the experiment is from UIEBD (underwater image enhancement benchmark dataset) dataset.³³ The dataset contains 950 real underwater images, 60 of which are challenging datasets with no reference images, and 890 with corresponding reference

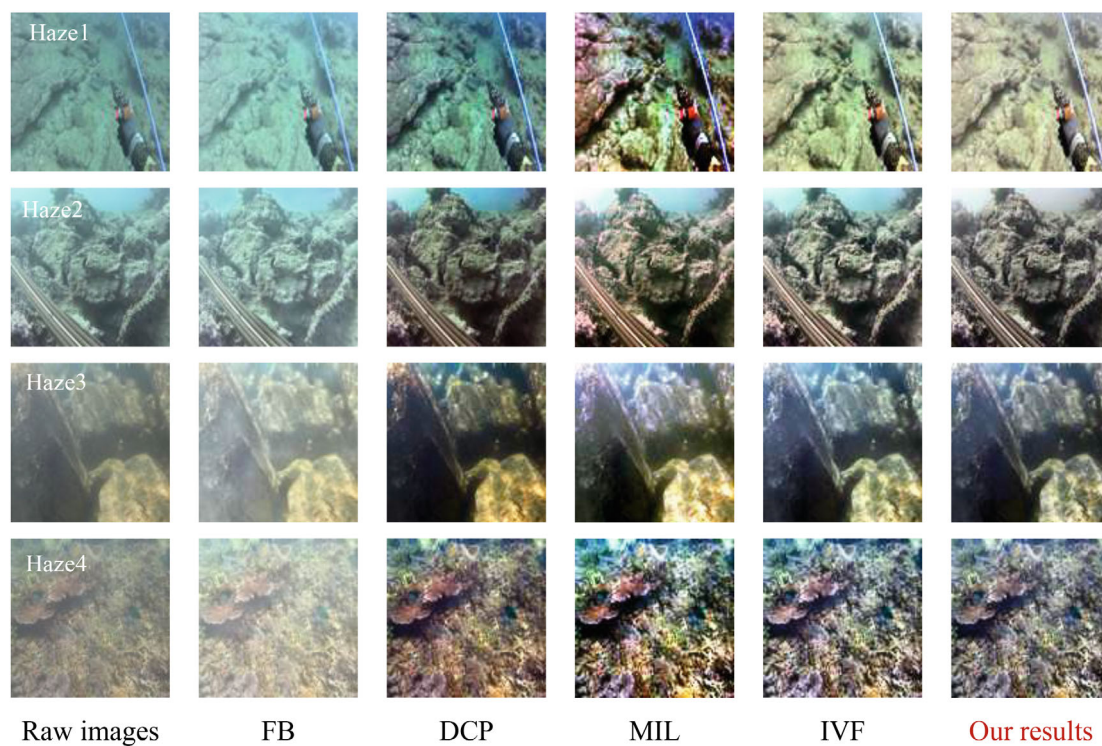


FIGURE 7 The haze underwater images

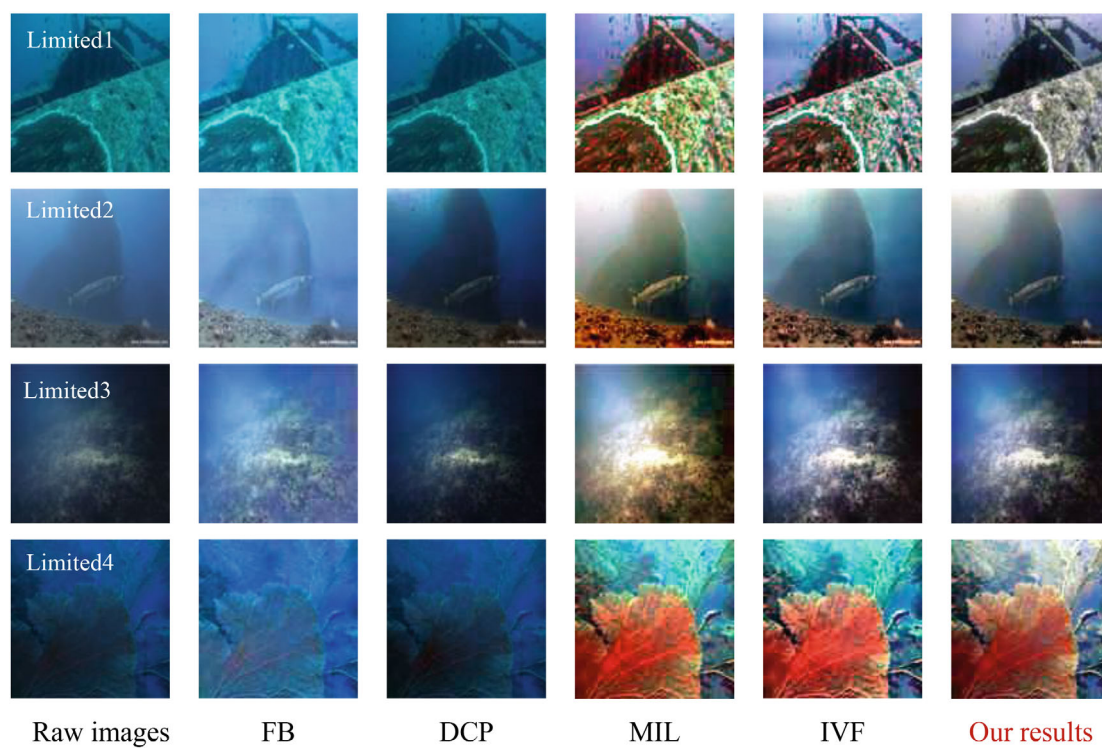


FIGURE 8 Underwater images with limited illumination

TABLE 1 PSNR, AG, IE, UIQM, and UCIQE values of different methods in Figure 5

Evaluation	Image	FB	DCP	MIL	IVF	Our method
PSNR	Green1	14.1275	8.4043	13.7008	14.4525	14.9758
	Green2	13.1793	8.4811	12.9585	14.7106	15.1323
	Green3	15.8015	9.6686	12.5708	14.1846	14.4435
	Green4	21.5980	7.6056	9.5843	10.1674	10.1250
IE	Green1	7.3081	7.2888	7.1552	7.8848	7.8253
	Green2	6.9501	7.3680	6.9964	7.8397	7.7580
	Green3	7.6976	7.3640	7.3568	7.8018	7.7076
	Green4	7.0155	7.3606	7.1621	7.8637	7.7886
UIQM	Green1	2.7680	4.8959	4.9651	4.0048	4.6984
	Green2	1.9700	4.2577	5.3080	4.9809	4.9325
	Green3	1.4217	2.9876	5.1507	5.0560	5.4610
	Green4	0.0956	1.2788	4.8962	4.6688	4.9322
UCIQE	Green1	0.4209	0.5259	0.7288	0.4233	0.6134
	Green2	0.3464	0.5582	0.6926	0.3658	0.6149
	Green3	0.3962	0.4840	0.6594	0.4303	0.5974
	Green4	0.3351	0.4166	0.7110	0.3893	0.6325

images, with resolutions ranging from 183×275 to 1350×1800 . The computer used in the experiment is configured with CPU Intel(R) Core(TM) i7-6700 3.4GHz, RAM 16 GB. Experimental algorithm programming using Matlab R2014b.

3.1 | Subjective evaluation

In order to illustrate the effectiveness of the algorithm in this article, several underwater images were selected from UIEBD, and then the images were divided into four categories: greenish images, bluish image, haze water images, and underwater image with limited illumination. The results of different methods are shown in Figures 5–8. Note that these underwater images cannot cover the total UIEB.

For the greenish images shown in Figure 5, FB and DCP methods could not correct the color bias of the image and the image visibility was low. Compared with IVF, FB, and DCP methods improved the green bias of the image. The green1 red channel in MIL method was overcompensated, while IVF method still had a certain color bias. The image produced by this method is superior to the four contrast algorithms in color, visibility and detail.

For the bluish images shown in Figure 6, FB and DCP methods were poor in improving image color and enhancing contrast, MIL and IVF methods improved image contrast, but red artifacts were introduced in blue1, blue2, and blue3. The visual effect of this method is satisfactory.

For the haze images shown in Figure 7, FB method did not improve the visual effect of the image, but reduced the contrast of the image. DCP method eliminated the influence of haze on the image and improved the contrast and clarity of the image. However, in haze1, the image had a certain colorcast, while MIL method appeared distortion, such as haze4. It also improves the visibility of the image. The proposed method is superior to IVF methods in improving image color, such as haze1 and haze2, and enhances image contrast and detail.

For the underwater images with limited illumination shown in Figure 8, FB method improved the brightness of the image, FB and DCP method failed to improve the color deviation of the image, MIL and IVF method introduced red artifacts, such as limited illumination1, limited illumination3 of MIL method appeared overexposure phenomenon. The proposed method achieves good visual effects in color correction, brightness and contrast.

TABLE 2 PSNR, AG, IE, UIQM, and UCIQE values of different methods in Figure 6

Evaluation	Image	FB	DCP	MIL	IFV	Our method
PSNR	Blue1	17.2263	6.3331	10.3469	9.6004	10.8993
	Blue2	18.1142	6.5801	9.0921	9.8303	10.7567
	Blue3	20.5494	6.3323	8.7789	10.3913	10.9150
	Blue4	16.3083	5.7177	9.9254	11.1307	10.8872
IE	Blue1	7.3575	7.5870	7.2880	7.8920	7.7757
	Blue2	6.6050	6.7191	6.7632	7.6592	7.5849
	Blue3	6.9293	6.8623	6.7879	7.5053	7.7117
	Blue4	6.8059	7.1105	6.9784	7.8710	7.8120
UIQM	Blue1	0.3346	1.4235	4.0690	4.5218	4.3120
	Blue2	−0.2189	0.7568	6.1312	4.1563	5.3761
	Blue3	0.4889	1.1191	5.8854	3.4472	4.9563
	Blue4	0.0712	1.3545	4.5178	4.4585	4.8372
UCIQE	Blue1	0.3840	0.4752	0.6593	0.4851	0.5729
	Blue2	0.3660	0.4255	0.6521	0.4255	0.5660
	Blue3	0.4402	0.4390	0.6404	0.4390	0.6005
	Blue4	0.2771	0.3538	0.6221	0.3153	0.5259

TABLE 3 PSNR, AG, IE, UIQM, and UCIQE values of different methods in Figure 7

Evaluation	Image	FB	DCP	MIL	IFV	Our method
PSNR	Haze1	17.7141	6.1721	12.5032	16.1851	15.5021
	Haze2	20.2114	7.5744	15.3545	17.1142	16.0822
	Haze3	14.2751	9.0588	15.8216	17.4042	16.5109
	Haze4	14.6252	6.6730	12.9741	15.1957	16.3954
IE	Haze1	7.0176	7.3773	7.0773	7.6380	7.3367
	Haze2	7.2210	7.6150	7.7196	7.8743	7.8429
	Haze3	6.7111	7.4076	7.5017	7.7925	7.6814
	Haze4	6.5674	7.4638	6.9700	7.8299	7.6744
UIQM	Haze1	3.3412	4.4355	4.8145	4.9460	5.2877
	Haze2	4.4108	5.5522	5.0277	5.1441	5.3867
	Haze3	4.2897	5.7581	5.4553	5.3017	5.5168
	Haze4	4.9810	6.0708	4.4549	5.0256	5.3313
UCIQE	Haze1	0.4015	0.5150	0.6394	0.3922	0.5205
	Haze2	0.4274	0.5882	0.6161	0.4748	0.5525
	Haze3	0.3978	0.6170	0.6310	0.4352	0.5820
	Haze4	0.3692	0.5835	0.6524	0.3742	0.5751

TABLE 4 PSNR, AG, IE, UIQM, and UCIQE values of different methods in Figure 8

Evaluation	Image	FB	DCP	MIL	IFV	Our method
PSNR	Limit1	15.0008	8.7044	10.4029	10.8020	11.8338
	Limit2	14.5281	11.3001	13.4767	18.0877	16.4591
	Limit3	9.8687	14.0152	10.5227	13.1132	15.4053
	Limit4	13.1290	12.0193	9.2831	8.8704	8.8993
IE	Limit1	7.5601	7.2121	7.0646	7.9255	7.7684
	Limit2	7.3607	7.0160	7.4448	7.7469	7.7140
	Limit3	7.5076	6.6028	7.2317	7.7401	7.6233
	Limit4	7.5804	6.9265	6.9319	7.9283	7.8904
UIQM	Limit1	0.5742	1.7980	4.9686	4.2995	5.0916
	Limit2	2.2710	4.1942	4.0634	3.5417	4.0139
	Limit3	3.3561	4.6752	4.8145	4.8680	4.9161
	Limit4	2.0389	3.5499	5.2351	4.8985	5.7991
UCIQE	Limit1	0.4427	0.4544	0.6764	0.4563	0.5902
	Limit2	0.4123	0.5688	0.6712	0.4740	0.6224
	Limit3	0.4449	0.5104	0.6568	0.4690	0.6199
	Limit4	0.4116	0.4587	0.7349	0.4453	0.6997

3.2 | Objective evaluation

Compared with subjective evaluation, objective evaluation is more stable and persuasive. Furthermore, we select two full-reference metrics includes peak signal to noise ratio (PSNR) and information entropy (IE) to evaluate the quality of underwater images shown in Figures 5–8. In addition, we further select two non-reference metrics includes underwater image quality measure (UIQM) and underwater color image quality evaluation metric (UCIQE) to evaluate the quality of underwater images shown in Figures 5–8.

PSNR is the mean square error between the original image and the processed image relative to the distortion degree of the represented image. The larger the value is, the higher the retained image quality information is, and the smaller the distortion is. IE reflects the amount of information contained in the image, IE bigger, more rich information, texture is clearer. UIQM represents the chrominance, sharpness and contrast of the image, and the larger the value, the better the image effect. The higher the UCIQE, the better the image is in color, saturation, and contrast.

Tables 1–4 given the evaluation scores of these methods applied to greenish, bluish, haze, and underwater images with limited illumination shown in Figures 5–8 in terms of PSNR, AG, IE, UIQM, and UCIQE. In the table, the first and second performance values of each indicator are bolded. In terms of PSNR, the image enhanced by FB method still has a large blue or green colorcast similar to the original image, which affects the PSNR results.^{35,36} The PSNR value of enhanced images by DCP and MIL methods is relatively low, and the PSNR value of IFV method is not much different from that of the method in this article. However, red artifacts are introduced into images processed by IFV method, as shown in blue1–4 and limit1. The method in this article has achieved good results for images of different scenes. In terms of IE, FB, DCP, and MIL methods had low values of enhanced images, while IVF method had the highest values, indicating that enhanced images contained large information and clear texture. However, red artifacts appeared in blue1–4 and limit1, and the IE value of this method was slightly lower than IVF method. In Figures 5–8, the image processed by this method has a good overall effect. In terms of UIQM, on the whole, the UIQM value of the proposed method is higher than that of other methods, indicating that the image processed by the proposed method has better color, sharpness and contrast. In terms of UCIQE, FB, DCP, and IFV methods have smaller values, while MIL method has the largest value, indicating that the saturation and contrast of enhanced images are high. However, red artifacts appear in blue1–4 and limit1, and overexposure occurs in limit3. The UCIQE value of this method is slightly lower than that of MIL method. The enhanced image achieved better results. To sum up, the proposed method achieves good visual effects for different types of images.

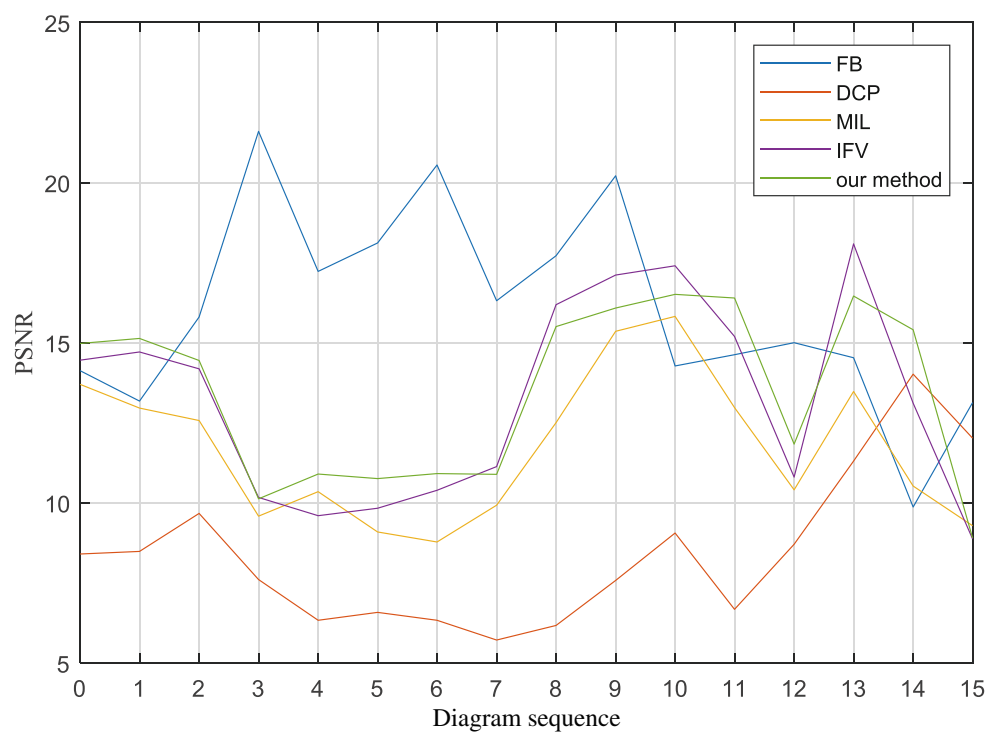


FIGURE 9 PSNR line chart

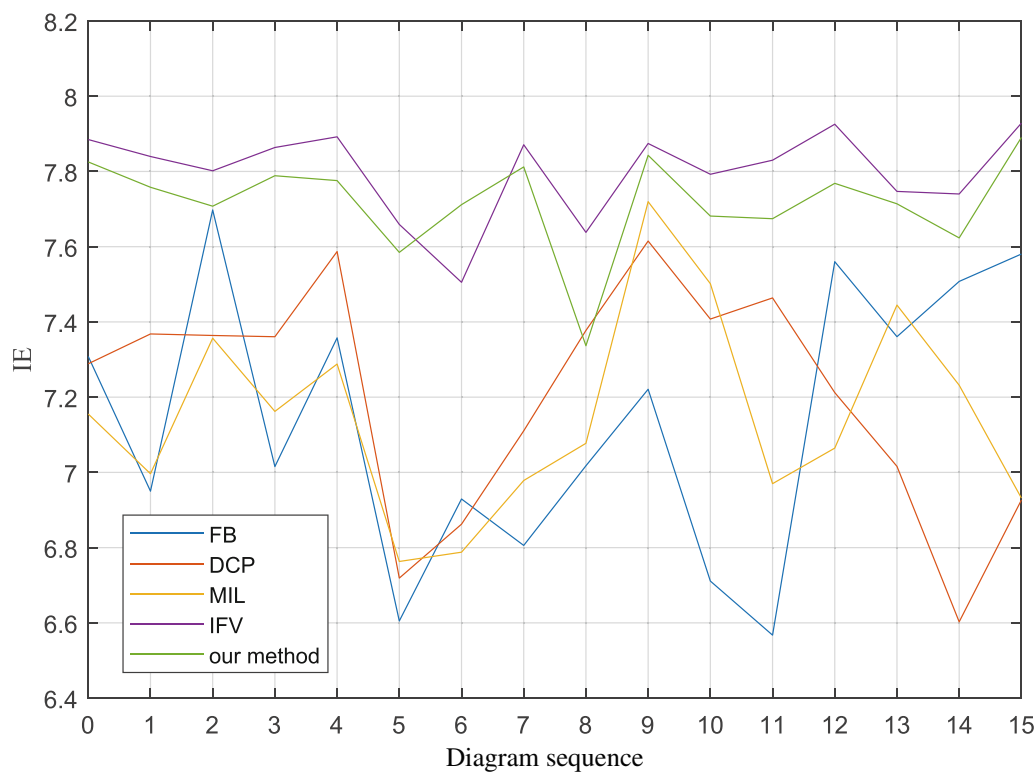


FIGURE 10 IE line chart

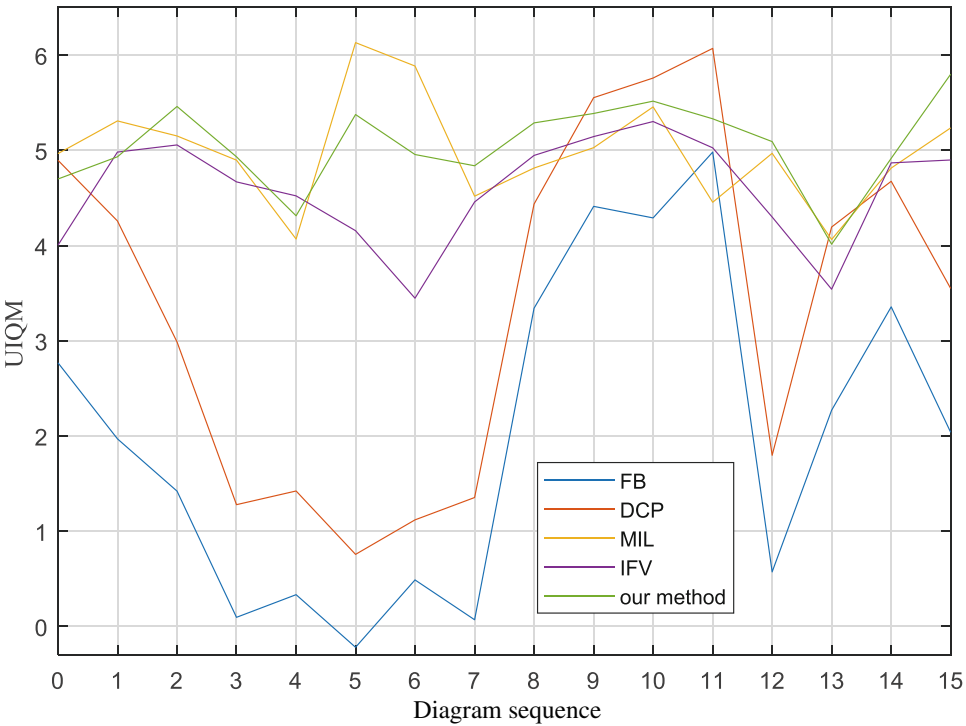


FIGURE 11 UIQM line chart

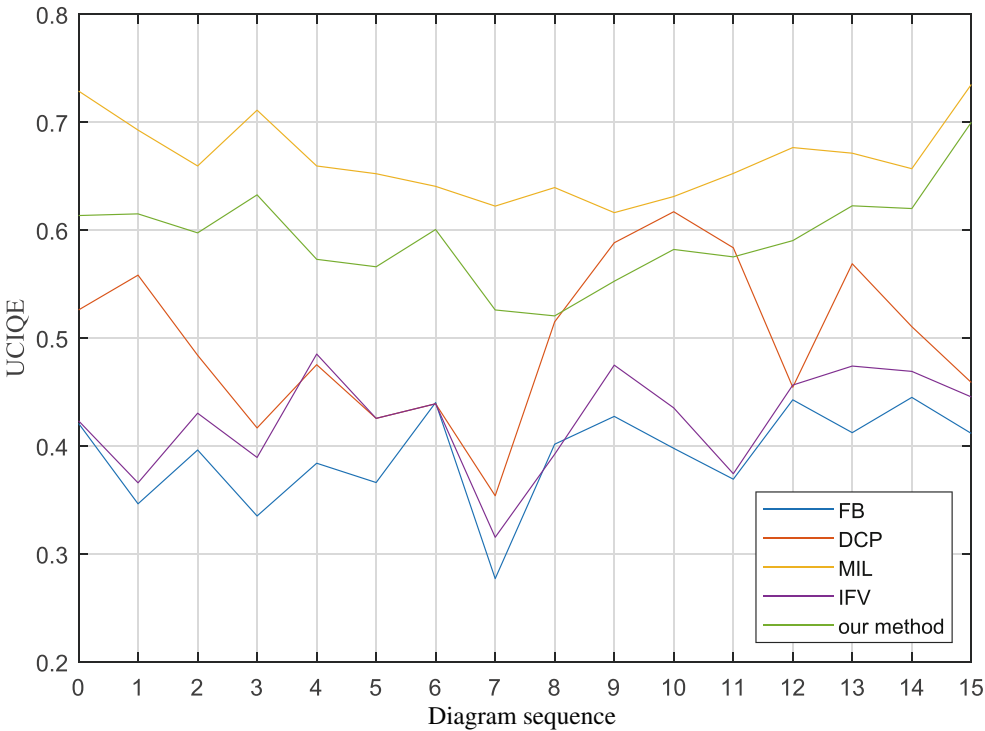


FIGURE 12 UCIQE line chart

In order to further analyze the objective indicators of underwater images, in this article, the images in Figure 5 are numbered as 0–3, the images in Figure 6 are numbered as 4–7, the images in Figure 7 are numbered as 8–11, and the images in Figure 8 are numbered as 12–15. The data in Tables 1–4 corresponding to Figure 5–8 are drawn into a line chart, as shown in Figures 9–12.

For Figure 9, the PSNR value of FB enhanced image is large, but the color deviation of the image is not improved. The PSNR value of DCP and MIL methods is low, and the value of FV method and the method in this article are high and not different, but the image enhanced by FV method introduces red artifacts, as shown in blue1–4 and limit1. The enhanced image distortion degree is small. As for Figure 10, it can be clearly seen that the IE values of FB, DCP, and MIL methods are low, and the IE values of IFV method are slightly higher than that of this method. However, the enhanced images of IFV method introduce red artifacts, as shown in Figure 6 and Limit1 in Figure 8. As for Figure 11, the UIQM values of FB and DCP methods vary greatly in the image processing of different scenes and are not universal. The UIQM value of IFV method is slightly lower than that of the method in this article, and the UIQM value of MIL method is not different from that of the method in this article, but the MIL method introduces red artifacts in blue1–4 and limit1. Therefore, the enhanced image color and contrast effect are better. For Figure 12, FB and IFV methods have small values, while DCP method has great changes for different images. MIL method has slightly higher values than the method in this article, but blue1–4 and limit1 have red artifacts, so the method in this article has a good effect on image enhancement. In general, the proposed method achieves good results for different types of images.

4 | CONCLUSIONS

Due to the special underwater imaging environment, underwater image color distortion, contrast decline and detail blur are some problems. Based on entropy weight fusion framework, an underwater image enhancement method based on multi-space transformation is proposed. First, the loss of the red channel is compensated, and then the color deviation of the image is corrected. The color deviation corrected image is processed in the lab color space to improve the difference between the brightness and darkness of the image; the image is processed in RGB color space to improve image contrast and remove image noise; process the image in HSV color space to improve the detail information and color information of the image. Finally, the obtained results are converted back to RGB color space for entropy weight fusion to obtain the enhanced image. Subjectively, the proposed algorithm has good color effect and clarity. Objectively, under the same conditions, compared with the contrast algorithm, the image enhancement effect of the algorithm presented in this article is better, and it achieves good results in PSNR, IE, UIQM, and UCIQE. In summary, the proposed algorithm achieves good enhancement effect.

ACKNOWLEDGMENTS

The authors very much appreciate the support to this research by: (1) The Key Science and Technology Innovation Programs in Shandong Province (NO.2017CXGC0919), and (2) the Shandong Province Key Research and Development Plan (No.2016GSF201197).

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How to cite this article: Du X, Sun X, Wang K, Yang J, Wang C. Underwater image enhancement method based on entropy weight fusion. *Comput Anim Virtual Worlds*. 2023;34(2):e2098. <https://doi.org/10.1002/cav.2098>